

UNIT-3rd Digital Logic Gate Characteristics

Basic Needs

Semiconductor devices :- 2 types $\begin{cases} \text{Bipolar} \rightarrow \\ \text{unipolar} \end{cases}$

Bipolar $\rightarrow$ Both types of charge carrier

$\wedge$
Electrons & Holes

Unipolar $\rightarrow$ Only one type of charge carrier :-
i.e. flows the current.

we use here switching property of this :-

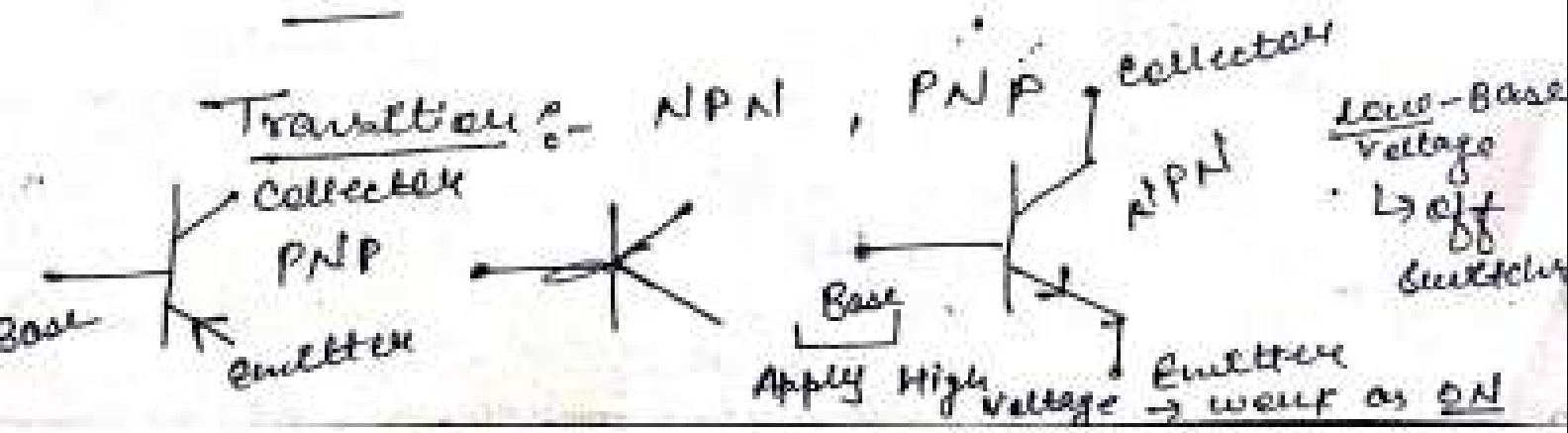
Diode $\rightarrow$ PN Junction Diode



Anode $\rightarrow$ unidirectional property follows

F.B $\rightarrow$ P $\rightarrow$ Anode $\rightarrow$ +ve Terminal of Battery
 $\rightarrow$ N $\rightarrow$ Cathode $\rightarrow$ -ve Terminal of Battery
 $\rightarrow$ so it works as ON switch

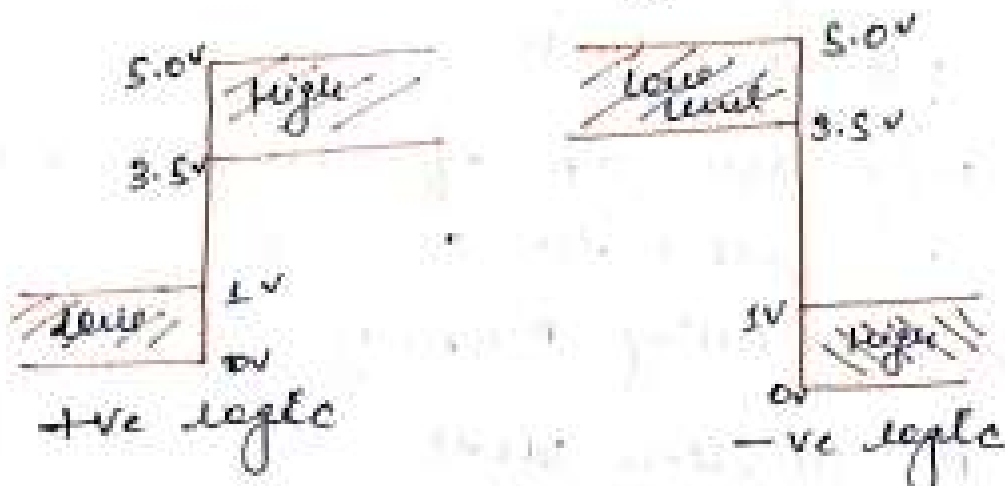
R.B $\rightarrow$ OFF switch.



Logic Family :- A group of compatible ICs with same logic level and supply voltage fabricated for performing various logic functions referred as a logic family. $\rightarrow$ OR, NOT, AND, etc.

IC $\rightarrow$ Integrated Circuit $\rightarrow$ chip

1000 of Transistors fabricated of that chip.



These levels are different for different logic families.

CMOS: Complementary Metal Oxide Semiconductor Field Effect Transistor

Digital IC

$\rightarrow$ I/P $\rightarrow$ logic OPⁿ CMOS
 $\rightarrow$ O/P $\rightarrow$ Bipolar

Bipolar logic family

$\rightarrow$ (use both electrons & holes as a charge carriers)

$\rightarrow$ register, Transistor, Diode are mainly used in Bipolar logic family.

MOSFET logic family

$\rightarrow$ (use only one type of charge carriers i.e. Holes or E.)

$\rightarrow$ MOS devices

- 1) PMOS - P-channel MOSFET
- 2) NMOS - N-channel MOSFET
- 3) CMOS - C-channel MOSFET

Saturated $\rightarrow$ IIL, HTL, RTL, DTL, DCTL, TTL, Non-saturated $\rightarrow$ ECL

$\rightarrow$ only work in Active or cut-off Region

Characteristics of Digital ICs :-

- 1) Propagation Delay, Speed
- 2) Power Dissipation
- 3) Figure of Merit
- 4) Fan-out, Fan-in
- 5) Noise Immunity, Noise Margin
- 6) Operating Temp.
- 7) Current and Voltage Parameters.
- 8) Breadth, Logic Swing

Propagation Delay :-

* Propagation delay is always measured by nano seconds, $\rightarrow 10^{-9}$ of a second

If TTL IC :- $t_{PHL} = 7ns$

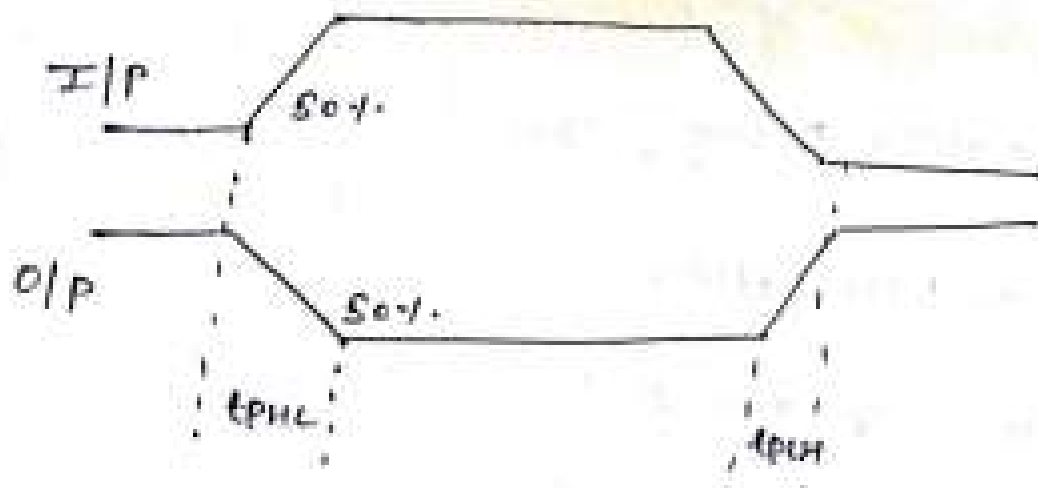
$$t_{PLH} = 11ns$$

Avg Prop Delay = ?

$$= \frac{11+7}{2} = 9 \underline{ns}$$

→ Speed of operations is specified in terms of Propagation delay.

→ Delay time is measured b/w 50% of IP + O/P Transition.



Avg. Propagation delay = $\frac{t_{PHL} + t_{PLH}}{2}$

t_{PHL} :- Delay time when O/P goes from High to Low

t_{PLH} :- Delay time when O/P goes from Low to High



→ Take some time → i.e. → Delay

Power Dissipation:-

→ This is the amount of Power dissipation is an IC in the form of Heat.

→ Accessing temp. could damage the IC, so power dissipation should be less.

→ P.D is determined by the current I_{CC} that is drawn from V_{CC}

$$P.D = I_{CC} \times V_{CC}$$

Figure of Merit :-

It is always define as a product of Speed + Power.

$$\text{Figure of Merit} = \text{Propagation delay} \times \text{Power}$$

$$\rightarrow \text{ns} \times \text{mW}$$

$$\rightarrow 10^{-9} \text{ sec} \times 10^{-3} \text{ W}$$

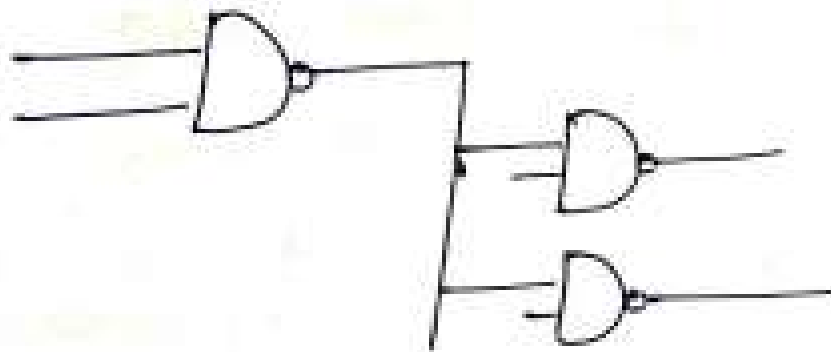
$$\rightarrow 10^{-12} \text{ Joules or Picoseconds}$$

$$\Rightarrow 10^{-12} \text{ PJ}$$

Fan-Out :-

No. of similar gates which can be driven by a Gate.

$\rightarrow$ Higher the fan-out, higher the current supplying capacity of a gate.



To other I/Ps

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Voltage Parameters :-

I_{IL}, I_{OL}	Current flow { { { {	Low level I/P	} } } }	Voltage
V_{IL}, V_{OL}		High level I/P		
		Low level O/P		
		High level O/P		

Noise Margin :- It is the limit of Noise Margin that can be present without disturbing proper operation of the circuit.

The circuit ability to tolerate Noise signal is referred to Noise Immunity.

Operating Temp :- at which Temp IC's operates.

0 to 70°C → Commercial Purpose

-55°C to 125°C → Military Application

RTL

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Register Transfer Logic

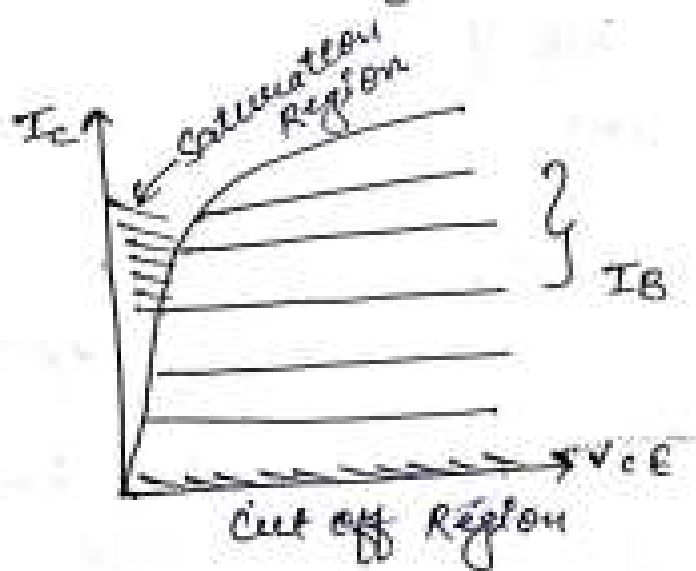
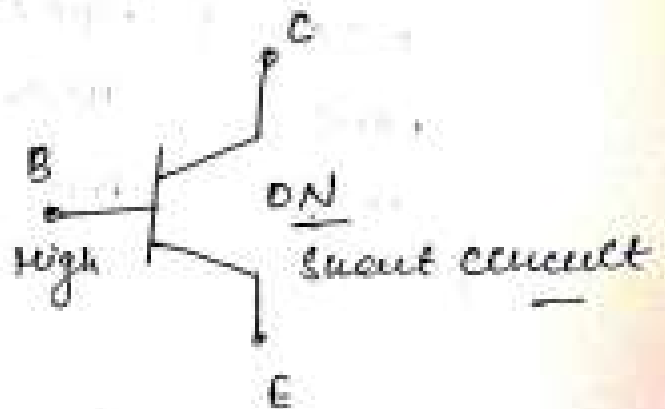
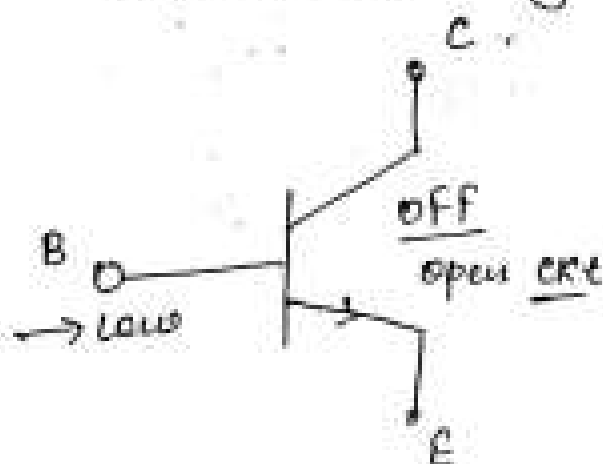
In this we use Register & Transfer to design RTL logic. OR we use new NOR gate.
↳ universal gate

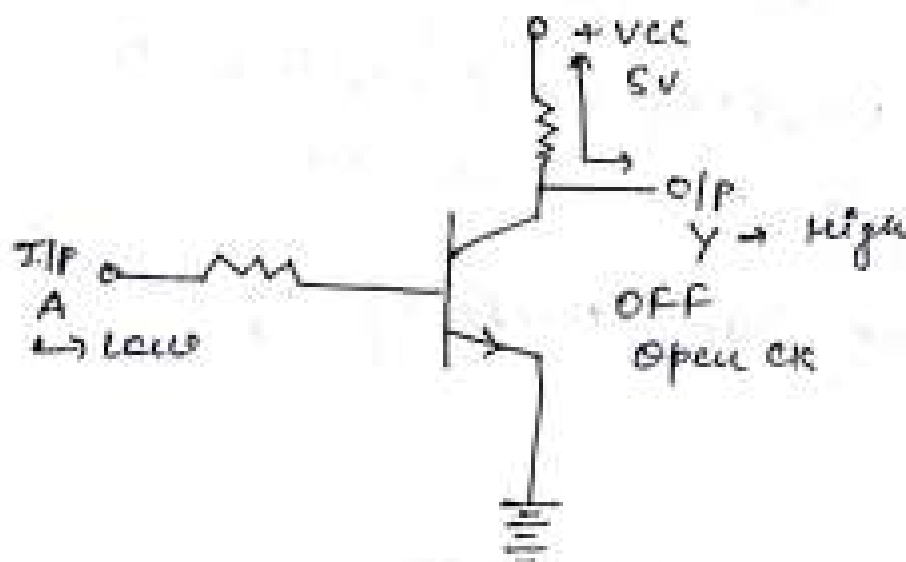
In this we use basic property of transistor
i.e. $\rightarrow$ ON

$\rightarrow$ OFF

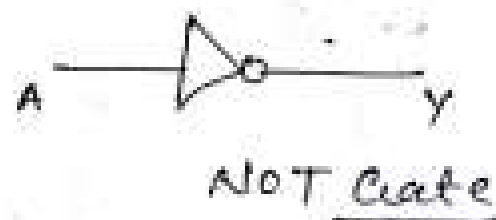
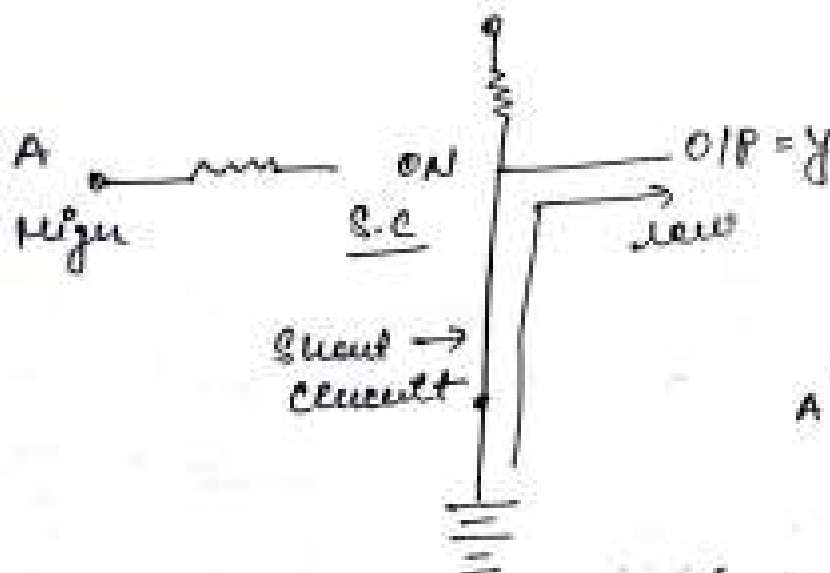
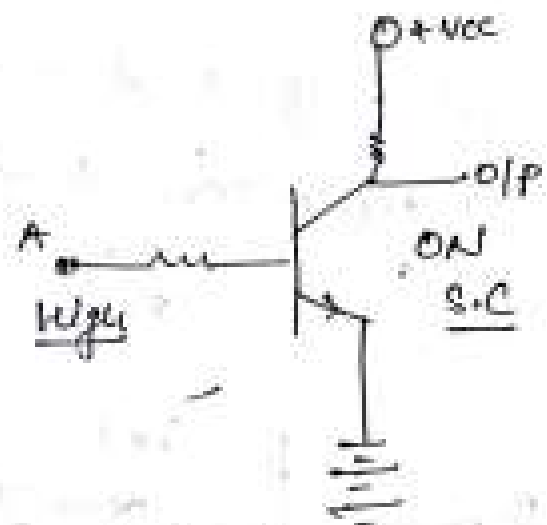
$\rightarrow$ Cut-Off Region $\rightarrow$ Off state $\rightarrow$ open circuit

$\rightarrow$ Saturation Region $\rightarrow$ ON state - S.C





I/P (A)	O/P (Y)
Low	High
High	Low



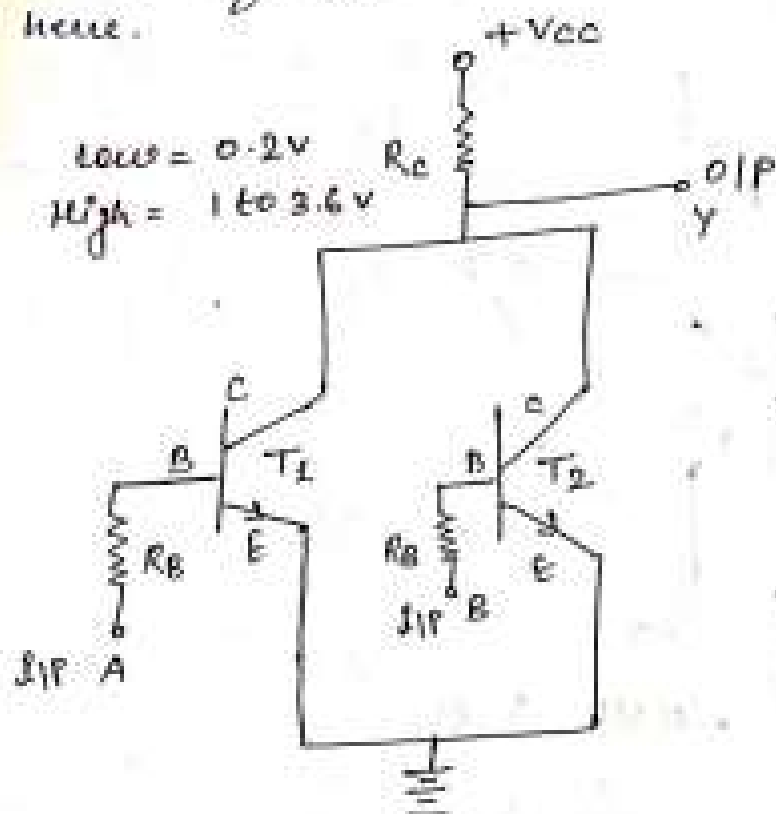
RTL Inverter Logic

Register Transistor Logic

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We also have
a no. of I/P
here.

Low = 0.2V
High = 1 to 3.6V

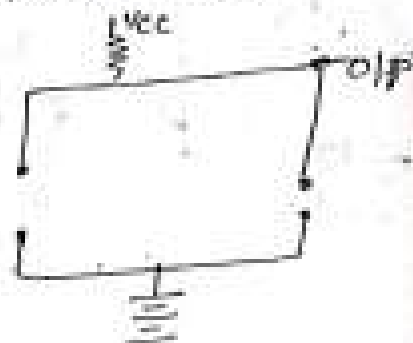


Basic gate - NOR

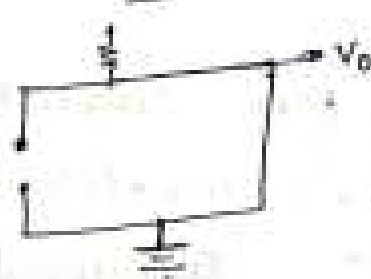
→ Base is connected
to I/P

→ Collector is connect
with V_{CC}

Case - 1



Case - 2



A	B	Y
Low	Low	High
Low	High	Low
High	Low	Low
High	High	Low

NOR



Diode Transistor Logic (DTL)

This is upgrade version of RTL in this
we improve different characteristics of
logic family. i.e.

Fan out, Power dissipation,
Noise Margin etc.

→ Low

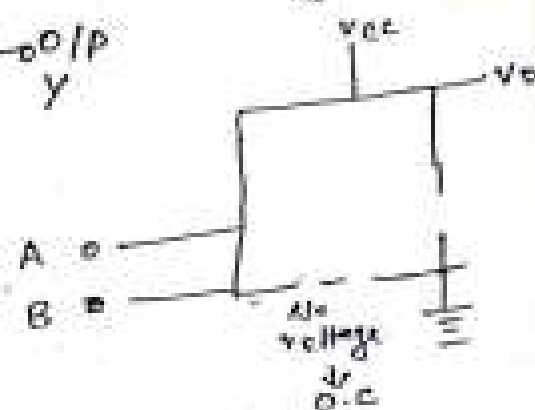
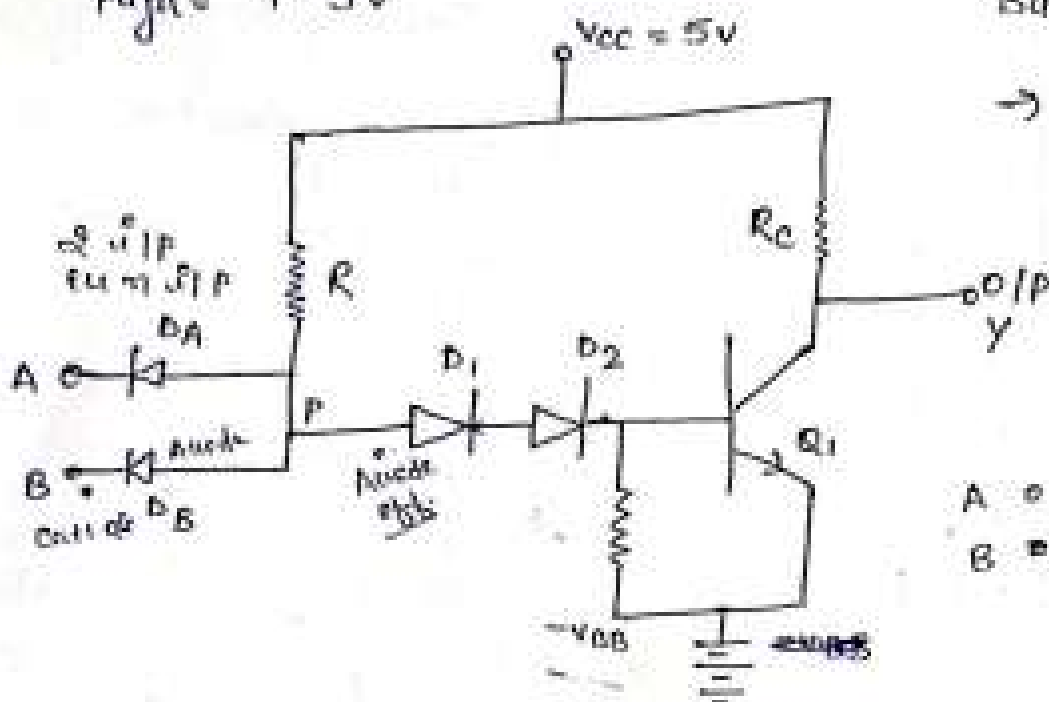
very high

→ We can design NAND
Gate using with
Transistor & Diode.

Low = 0.2V
High = 4-5V

Basic Gate $\rightarrow$ NAND

$\rightarrow$ Anode must be connected to higher voltage



$\rightarrow$ DTL circuit having NAND Gate.

$\rightarrow$ Greater Fan Out

$\rightarrow$ Improved Noise Margin

* If Transistor base is connected with -ve voltage means cut-off $\rightarrow$ off state $\rightarrow$ O.C

* Diode $\rightarrow$ ON $\rightarrow$ forward bias $\rightarrow$ current flow $\rightarrow$ S.C
OFF - Reverse bias $\rightarrow$ No current flow $\rightarrow$ O.C

Transistor $\rightarrow$ Base $\rightarrow$ low i/p $\rightarrow$ OFF $\rightarrow$ O.C

Base $\rightarrow$ High i/p $\rightarrow$ ON $\rightarrow$ S.C

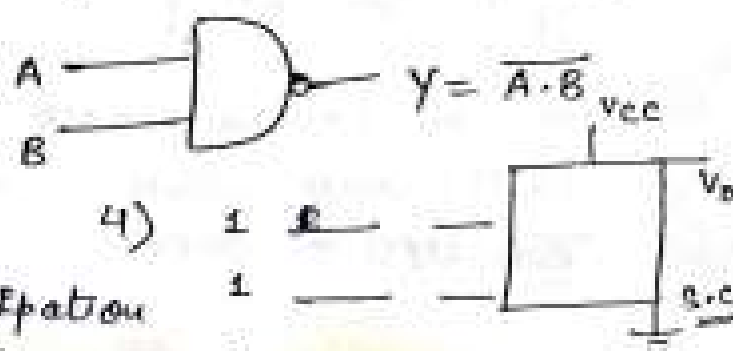
V_{ce} = saturation
low = 0.2V

A	B	Y
0	0	High (1)
0	1	High (1)
1	0	High (1)
1	1	low (0)

comple $\frac{1}{2}$
Bcoz of current

$\rightarrow$ Speed slow, Power Dissipation

$\rightarrow$ High Fan-out, Noise Margin $\rightarrow$ TTL Next

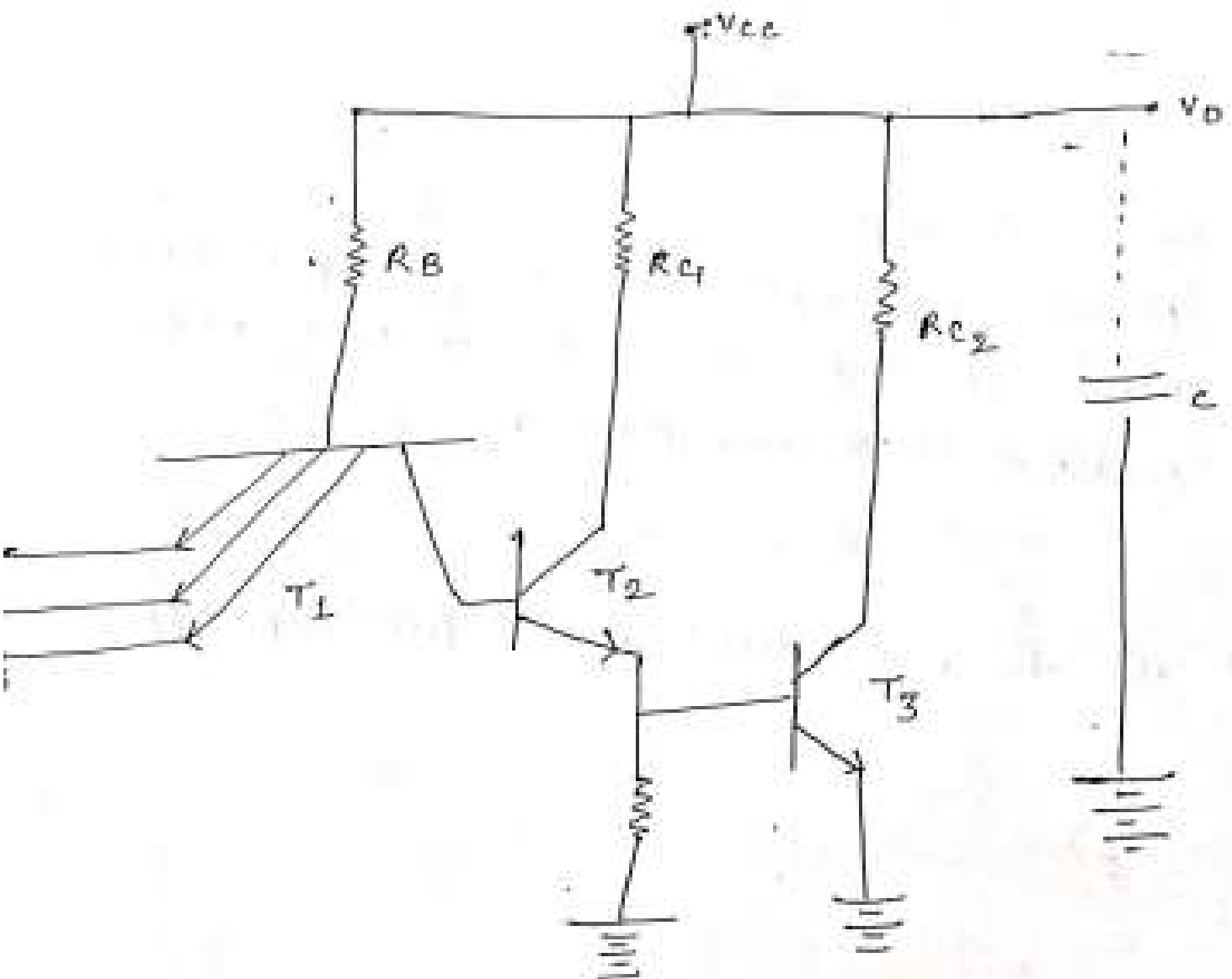


Transistor Transistor Logic

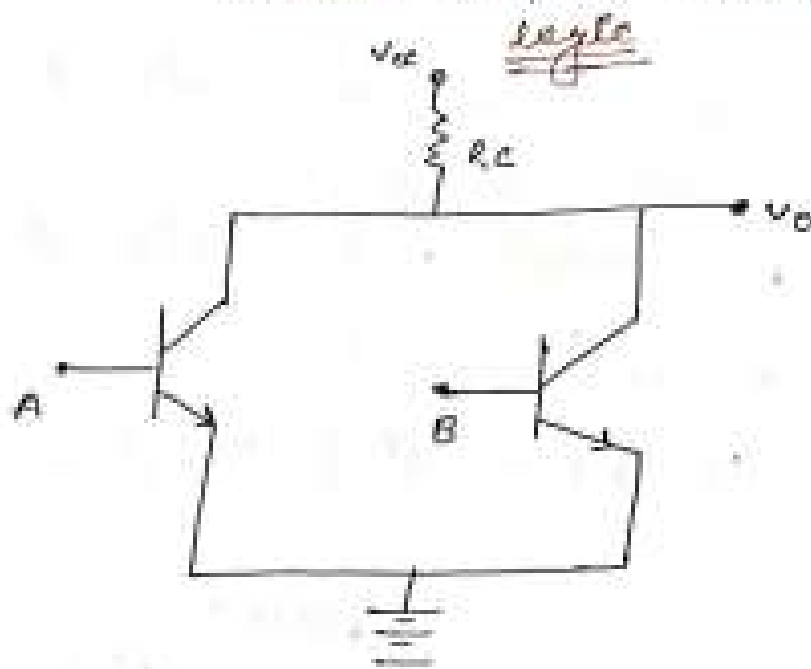
11.

TTL

- This is just modified circuit of RTL.
- Here we are using few V_{BE} in multi-emitter Transistor.
- In this we remove the Diode - D_1 + D_2 with the help of Transistor.



Directly Coupled Transistor



Basic gate - NOR
 In RTL :-
 → Poor Noise Margin
 → High Power Consumption

- In this SIP directly connected to base of Transistor without using any Register in between Input & Transistor Base.
- functionality is similar to RTL, so we use NOR Gate.

- The SIP of Transistor are low, then transistor in cut-off region → O.C ^{Input} _{to output}
- The SIP of Transistor are high, then transistor in saturation region → S.C

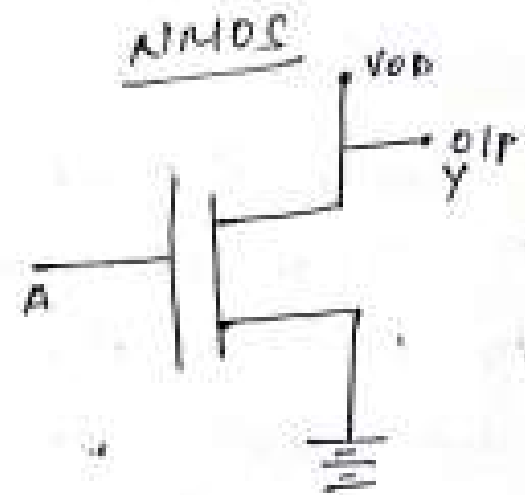
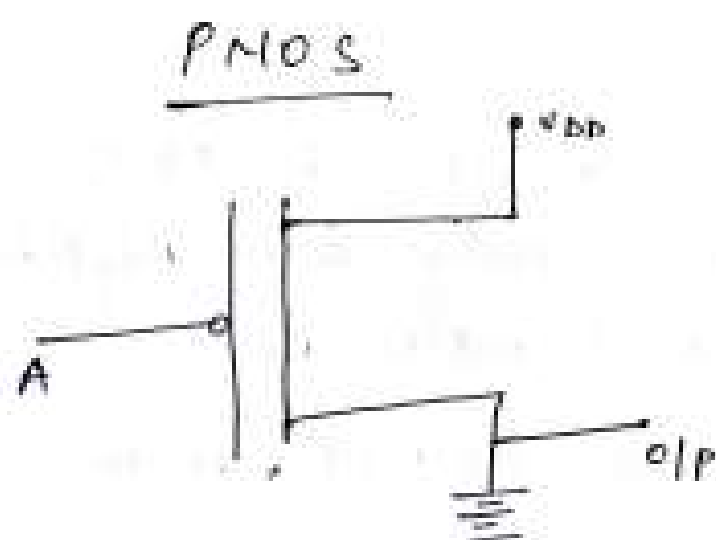
SIP		SIP
A	B	V _O
0	0	1
0	1 → ON	0
1	0	0
1	1	0



We use less no. of SIP.
 so we make a no. of NOR Gate.

PMOS & NMOS Transistors

- In this we use unipolar logic family.
- current direction is single.
- PMOS → P-channel Metal oxide Semiconductor
- NMOS → N-channel " " "
- CMOS → Complementary " " "



Voltage :-

0V (0)

5V (1)

PMOS

ON

OFF

NMOS

OFF

ON

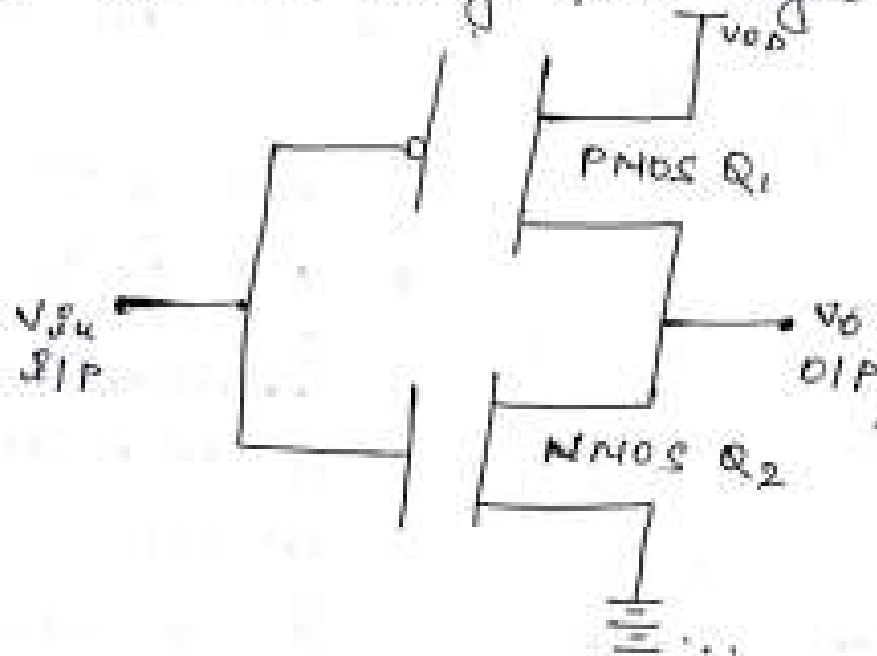
	SIP	O/P
	A	Y
ON C.C	0	1
OFF O.C → 1		0

	SIP	O/P
	A	Y
OFF O.C → 0		1
ON S.C → 1		0

CMOS Inverter

15.

- It is combination of PMOS & NMOS.
- It is very fast logic family.



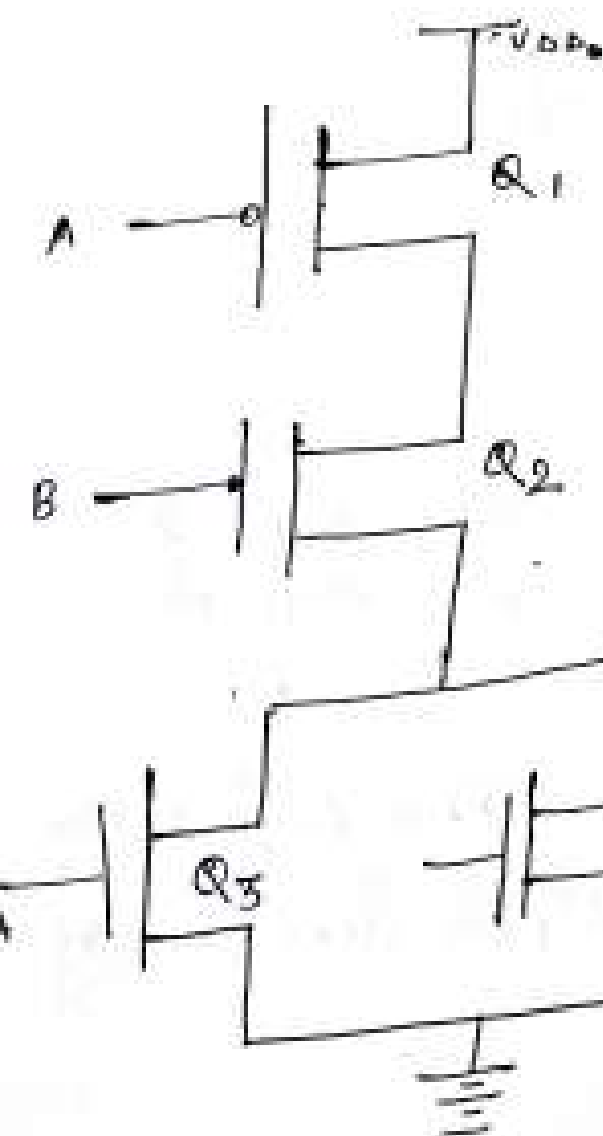
CMOS or CMOS Inverter is same as PMOS & NMOS or these Inverter is same.

- NMOS is going to ON for high voltage
 - PMOS is going to ON for low voltage
- | V_{in} | Q_1 | Q_2 | V_o |
|----------|-------|-------|-------|
| 0 | ON | OFF | 1 |
| 1 | OFF | ON | 0 |

- That's why CMOS also called CMOS Inverter.
- MSI, Highest speed logic family as compare to other family.

CMOS NOR Gate

→ CMOS NOR Gate use NMOS & PMOS
 use this functionally Full CMOS NOR Gate
 ↓ Pull down ↓ Pull up



A	B
0	0
0	1
1	0
1	1

PMOS
 ON - 0 → S.C
 OFF - 1 → O.C

PMOS	NMOS
Q1, Q2	Q3, Q4
ON ON	OFF OFF
ON OFF	OFF ON
OFF ON	ON OFF
OFF OFF	ON ON

This is work as NOR Gate operations.



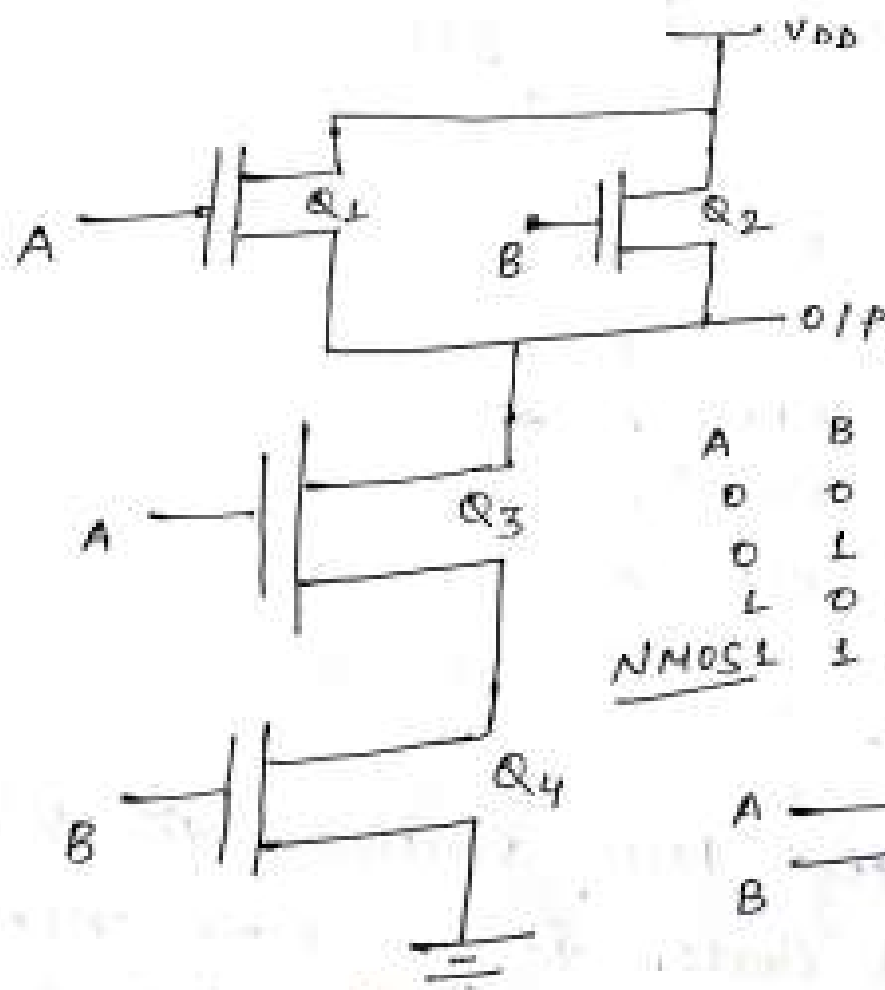
ON - 1 → S.C
 OFF - 0 → O.C

→ NMOS

CMOS NAND Gate

→ CMOS NAND Gate are designed by NMOS & PMOS logic family.

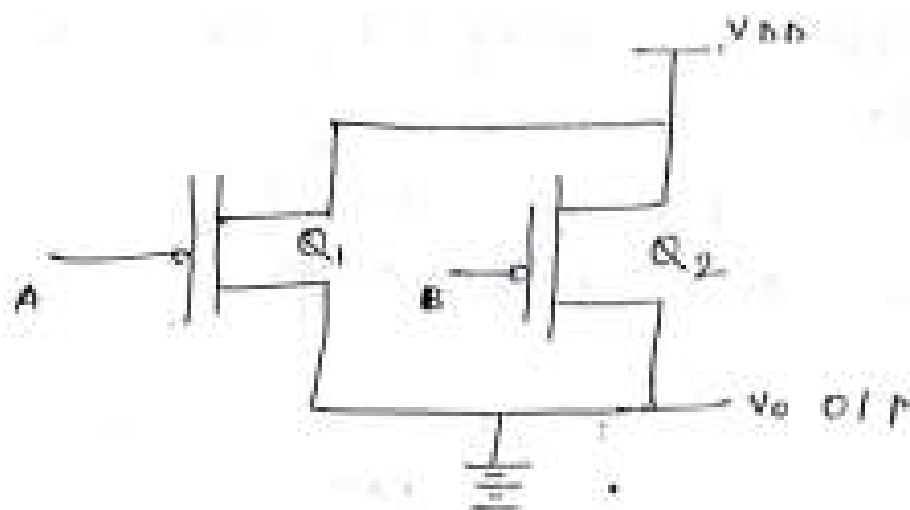
→ In this NMOS is pull up mode & PMOS is pull down mode, only.



A	B	P		N		V _O
		Q ₁	Q ₂	Q ₃	Q ₄	
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0



PMOS NAND Gate

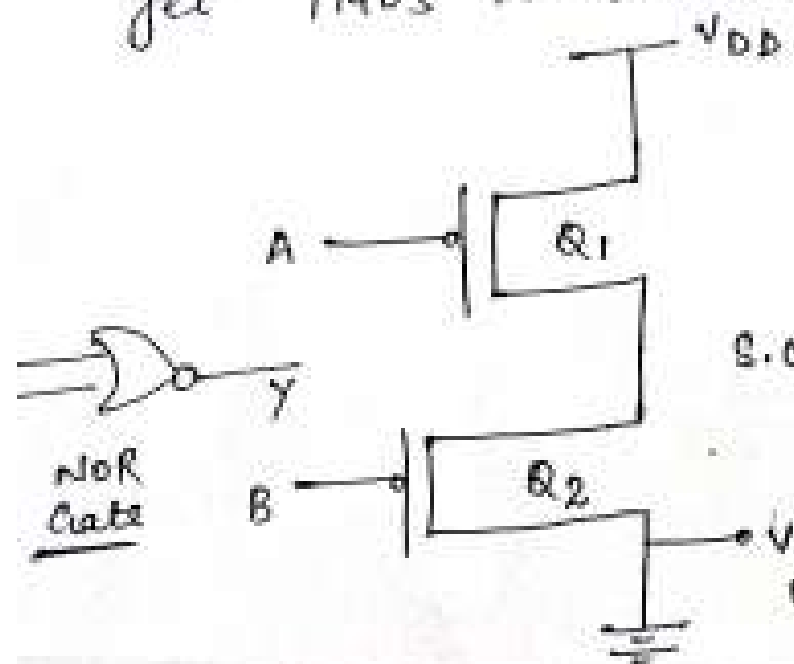


0 - ON - S.C
OFF → 1 - O.C

A	B	Q ₁	Q ₂	V _O
0	0	ON	ON	1
0	1	ON	OFF	1
1	0	OFF	ON	1
1	1	OFF	OFF	0

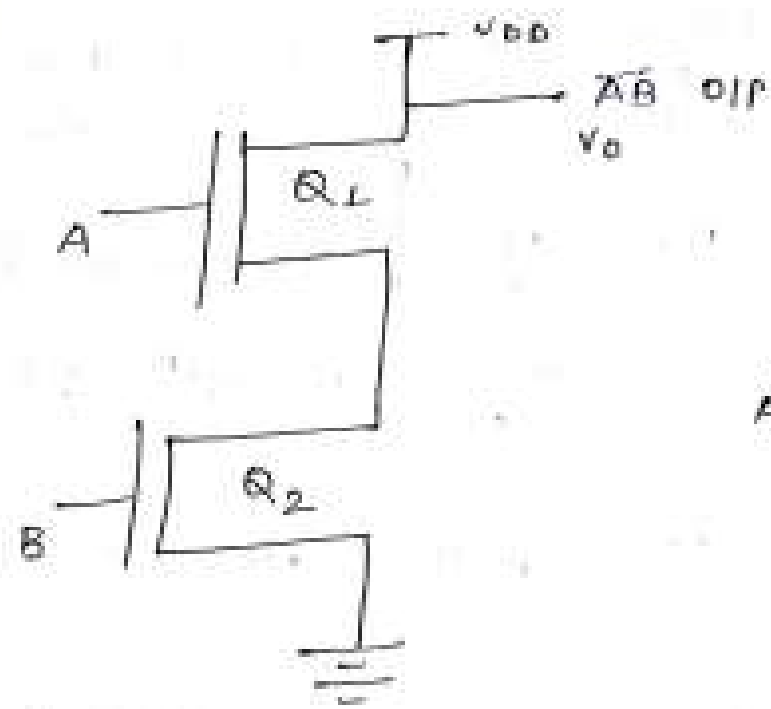
PMOS NOR Gate!

when we connected PMOS vertical then we get PMOS NOR Gate. if we want N no. of O/P then we add no. of I/P here like vertical.

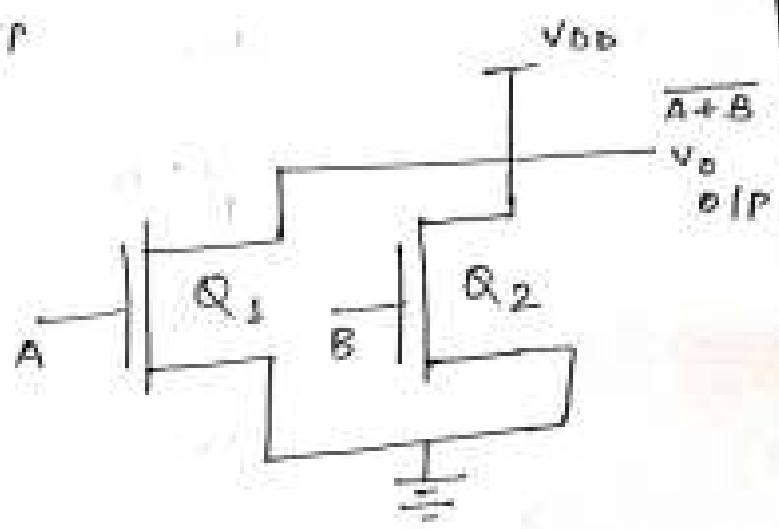


A	B	Q ₁	Q ₂	V _O
0	0	ON	ON	1
0	1	ON	OFF	0
1	0	OFF	ON	0
1	1	OFF	OFF	0

Designing Universal Gates using NMOS



NAND



NOR

A	B	Q ₁	Q ₂	V _O o/p
OFF → 0	0	OFF	OFF	1
↓				
0.c	0	OFF	ON	1
	1	ON	OFF	1
	0	ON	OFF	1
ON → 1	1	ON	ON	0
↓				
g.c				

A	B	Q ₁	Q ₂	V _O o/p
0	0	OFF	OFF	1
0	1	OFF	ON	0
1	0	ON	OFF	0
1	1	ON	ON	0
		↓		
		S.c		

Comparison of logic family 20.

	RTL	DTL	DTL	TTL	ECL	2^2 IL	CMOS
Parameter	Register & Transistor	Register	R, D, R, T	R, T	T	PMOS	NMOS
Components	Resistor	Transistor	T				
Noise - Margin	Poor	Poor	High	Medium	Low	High	High
Fan-out	Low (4)	Low (4)	Medium (8)	High (10)	High (15)	High (15)	High (15)
Power - dissipation (mW)	30	10	10	10	10	10	10
Delay (ns)	12	10	30	10	10	10	10
Figure of Merit	144	1800	300	100	100	100	100

CMOS is low power consumption.

ECL is fastest logic family